

# Christopher Musco

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## Education

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### Massachusetts Institute of Technology

*Ph.D. Candidate, Computer Science – Theory of Computation*

Advisors: Jonathan Kelner, Martin Rinard

**Cambridge, MA**

*2013 – present*

### Yale University

*B.S. Applied Mathematics, B.S. Computer Science*

AM Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

CS Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

**New Haven, CT**

*2008 – 2012*

## Research Interests

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I am interested in dimensionality reduction and its application to randomized numerical linear algebra, sketching and streaming algorithms, and machine learning. I also work on problems related to algorithmic graph theory and collaborate with MIT's Center for Resilient Software on quantitative approaches to computer security and approximate computation.

## Academic Positions

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### Massachusetts Institute of Technology

*Research Assistant*

**Cambridge, MA**

*2013 – present*

### Yahoo! Labs

*Research Intern, Scalable Machine Learning Group*

**New York, NY**

*Summer 2015*

## Teaching and Mentorship

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### Research Science Institute/MIT

*Summer Research Advisor*

*2014*

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Designed a project on concurrent data structures under the direction of Professor Martin Rinard.
- Chris' paper, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

### Yale University

*Teaching Assistant, Introduction to Computer Science*

*2012*

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

*Teaching Assistant, Introduction to Programming*

*2011*

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

*Undergraduate Grader, Introductory Calculus*

*2009 – 2010*

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

## Honors and Awards

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### National Science Foundation

*Graduate Research Fellowship*

*2014*

## Yale University

2012

*Cum Laude, Distinction in Both Majors*

## Publications

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### Conference Papers.....

[Dimensionality Reduction for K-Means Clustering and Low Rank Approximation](#). Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *To appear in STOC* 2015.

[Principled Sampling for Anomaly Detection](#). Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

[Uniform Sampling for Matrix Approximation](#). Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

[Single Pass Spectral Sparsification in Dynamic Streams](#). Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. (Invited to SICOMP Special Issue).

### Other.....

[Stronger and Faster Approximate Singular Value Decomposition via the Block Lanczos Method](#). Cameron Musco, Christopher Musco. *Preprint* 2015.

## Talks

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### **Dimensionality Reduction for K-Means Clustering**

Symposium on Theory of Computing (STOC) *Jun. 2015*

MIT, Theory Lunch *June 2014*

### **Principled Sampling for Anomaly Detection**

MIT, Defense Advanced Research Projects Agency Site Visit *Mar. 2015*

Network and Distributed System Security Symposium (NDSS) *Feb. 2015*

### **Single Pass Spectral Sparsification in Dynamic Streams**

MIT, Annual Sublinear Algorithms Day (Poster Session) *Apr. 2015*

Harvard University, TOC Seminar *Nov. 2014*

Foundations of Computer Science (FOCS) *Oct. 2014*

## Service

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### External Reviewer.....

European Symposia on Algorithms (ESA) *2015*

Foundations of Computer Science (FOCS) *2015*

International Colloquium on Automata, Languages and Programming (ICALP) *2015*

Network and Distributed System Security Symposium (NDSS) *2015*

## Professional Experience

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### **Redfin**

**Seattle, WA**

*Software Engineer, Data Team*

*2012 – 2014*

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code base.

**Elysium Digital***Summer Consultant*

Provided technical expertise in software patent cases at leading litigation consulting company.

**Cambridge, MA***Summer 2011***Amicus***Software Developer*

Built tools for political campaign management and social fundraising at early-stage startup.

**New Haven, CT***2010 – 2011***Moody's Investors Service***Intern, Asset Finance Group*

Established and monitored bond ratings on asset securitizations.

**New York, NY***Summer 2009***U.S. Department of Agriculture***Intern, Natural Resources Conservation Service*

Collected and analyzed data for state conservation and research projects.

**East Greenwich, RI***Summer 2007*